doi: 10.1111/obr.12503

Review

Role of sleep and circadian disruption on energy expenditure and in metabolic predisposition to human obesity and metabolic disease

A. W. McHill[1] and K. P. Wright Jr[2]

[1]Division of Sleep and Circadian Disorders, Department of Medicine, Brigham and Women's Hospital and Division of Sleep Medicine, Harvard Medical School, Boston, MA, USA, and [2]Sleep and Chronobiology Laboratory, Department of Integrative Physiology, University of Colorado Boulder, Boulder, CO, USA

Received 1 December 2016; accepted 2 December 2016

Address for correspondence: Dr AW McHill, Division of Sleep and Circadian Disorders, Department of Medicine, Brigham and Women's Hospital and Division of Sleep Medicine, Harvard Medical School, Boston, MA 02115, USA.
E-mail: amchill@bwh.harvard.edu
Dr KP Wright Jr, Sleep and Chronobiology Laboratory, Department of Integrative Physiology, University of Colorado Boulder, Boulder, CO 80309-0354, USA.
E-mail: kenneth.wright@colorado.edu

Summary

Weight gain, obesity and diabetes have reached alarming levels in the developed world. Traditional risk factors such as over-eating, poor nutritional choices and lack of exercise cannot fully account for the high prevalence of metabolic disease. This review paper examines the scientific evidence on two novel risk factors that contribute to dys-regulated metabolic physiology: sleep disruption and circadian misalignment. Specifically, fundamental relationships between energy metabolism and sleep and circadian rhythms and the impact of sleep and circadian disruption on metabolic physiology are examined. Millions of individuals worldwide do not obtain sufficient sleep for healthy metabolic function, and many participate in shift work and social activities at times when the internal physiological clock is promoting sleep. These behaviours predispose an individual for poor metabolic health by promoting excess caloric intake in response to reduced sleep, food intake at internal biological times when metabolic physiology is not prepared, decreased energy expenditure when wakefulness and sleep are initiated at incorrect internal biological times, and disrupted glucose metabolism during short sleep and circadian misalignment. In addition to the traditional risk factors of poor diet and exercise, disturbed sleep and circadian rhythms represent modifiable risk factors for prevention and treatment of metabolic disease and for promotion of healthy metabolism.

Keywords: Circadian misalignment, sleep fragmentation, sleep loss, sleep restriction.

Introduction

Over the last 2 decades, identifying behaviours that may promote weight gain, obesity and diabetes has come to the forefront of both public health and scientific interest. While the majority of research and public health efforts have focused on over-eating, unhealthy diet choices and inactivity as causes of adverse metabolic health, these factors cannot fully account for a continued high incidence of unwanted weight gain, obesity, diabetes and the metabolic syndrome in modern society (1). Thus, other potential risk factors must be identified in order to develop effective countermeasures to promote health and combat metabolic disease. Two such novel risk factors may be insufficient or disrupted sleep and circadian misalignment, the latter being when daily routines such as wakefulness and food intake occur when our internal circadian timing system is promoting sleep (2–4). A National Heath Survey reported that ~30% of Americans typically sleep less than 6 h per night (5), an amount below the recommended 7-h sleep duration for promoting optimal health in adults (6). Furthermore, access to electrical lighting has delayed the timing of the circadian clock and allowed for work and social activities to occur outside of typical daytime hours (7), with

© 2017 World Obesity Federation

~20% of the employed population working nontraditional worktime hours during the typical sleep time (8). These factors have been associated with risk of obesity and diabetes (9) and thus need to be examined as potential contributing mechanisms that may contribute to the rapid increase in poor health and are potential targets for intervention to improve metabolic health.

This review examines the existing literature pertaining to the relationship between sleep and circadian physiology with energy metabolism. It will first describe the fundamental integrations between sleep and circadian physiology with energy metabolism and then highlight the evidence for how and why disruption of sleep and circadian physiology contributes to disease. By recognizing these interactions and the outcomes of metabolic disturbance, this review will provide a foundation for future studies to expand upon to address these potentially modifiable factors in order to combat metabolic disease and promote health.

Sleep and energy metabolism

One of the proposed functions of sleep is energy conservation. In a seminal paper in 1983, Rechschaffen and collaborators used a rodent model to examine the effects of sleep deprivation on energy metabolism. The researchers deprived one rat of sleep while a 'yoked' control slept when the sleep-deprived rat was awake (e.g. feeding and grooming) (10). Experimental and yoked control rats were placed on a rotating disk above a small amount of water and were separated by a plexiglass wall. When the sleep-deprived rat entered sleep as determined via the electroencephalogram, the disk would rotate, and the sleeping rat would either awaken and begin walking to avoid being pushed into the water or fall into the water and be awakened. This procedure resulted in a loss of ~87% of sleep in the sleep-deprived rat compared with baseline. The yoked control rat was able to obtain ~70% of daily sleep compared with baseline. Total daily 24-h energy expenditure increased by ~60–100% and ~20–30% in sleep-deprived and the sleep-restricted yoked control groups compared with baseline, respectively (11,12). Furthermore, energy expenditure continued to increase across weeks of near total sleep deprivation. Although food and water were available *ad libitum* and the rats increased their total daily caloric intake, weight loss ensued in both groups indicating a high metabolic cost of sleep loss. The near totally sleep-deprived rats in particular exhibited a malnutrition phenotype and lost the ability to thermoregulate. These findings suggested that the sleep-deprived rats could not consume enough calories to meet additional energy requirements from sleep deprivation to maintain healthy metabolic physiology (11,12). Importantly, these findings provided early evidence that sleep deprivation increases energy expenditure more than does typical wakefulness in the rat model.

In humans, whole body energy expenditure has been shown to be lower during sleep than during wakefulness, as observed through decreased sleeping metabolic rate (13–15) and increased energy expenditure during sleep restriction (16–18) and total sleep deprivation (19). By studying participants under tightly controlled conditions in which seated bed rest was maintained and food intake was provided to meet daily energy requirements at baseline, 24-h total sleep deprivation was found to increase energy expenditure overnight by ~32% as compared with an 8-h night-time sleep opportunity and by ~7% over an entire 24 h d^{-1} (19). Similar findings have been found during sleep deprivation that also included a daytime nap (17). Following 40 h of total sleep deprivation, individuals were found to expend significantly less energy during a recovery sleep opportunity as compared with a baseline night of sleep at habitual timing (19). These findings suggest that the balance of energy consumed vs. expended is fluctuating around a homeostatically regulated set point and that sleep plays a role in maintaining energy balance. Following a night of total sleep deprivation, resting metabolic rate has also been reported to be reduced by ~5%, and the thermic effect of feeding, or the energy expended to digest and store food after consumption, was reduced by ~20% in the morning following the sleep deprivation (20).

The findings reviewed previously indicate a general consistency in the metabolic response in rodents and humans to total sleep deprivation; i.e. sleep deprivation increases energy expenditure (Fig. 1). Species differences are observed, however, with sleep deprivation appearing to be a bigger metabolic challenge to rodents as they show larger increases in energy expenditure. Additional research is needed to understand physiological mechanisms underlying such

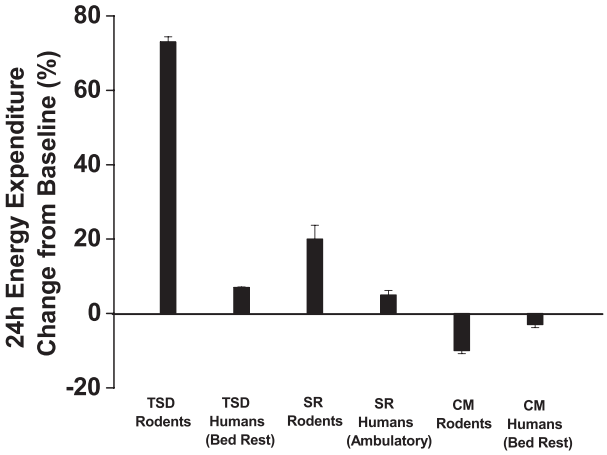


Figure 1 Change from baseline in energy expenditure after total sleep deprivation (TSD), sleep restriction (SR) and circadian misalignment (CM) in both rodent and human models. Energy expenditure has been found to be ~75% higher in rodents (110) and ~7% in humans (19) under TSD, ~20% in rodents (11,12) and ~5% in humans (16,18) under SR and ~10% lower in rodents (27) and ~3% in humans (17) under CM.

© 2017 World Obesity Federation

species differences in the metabolic cost of sleep loss (e.g. using omics techniques). With respect to the energy conservation function of sleep, we (19) and Schmidt (21) have proposed that energy utilization is repartitioned/reallocated to different physiological processes during sleep vs. wakefulness and understanding such partitioning, as well as what physiological processes contribute to increased energy expenditure during sleep loss, is an open area of research.

The circadian system and energy metabolism

The circadian system and energy metabolism are intimately coupled and have evolved to promote certain energetically costly activities either during the day or during the night. This can be observed in simple organisms and in complex physiological systems. In single-celled cyanobacteria, production of adenosine triphosphate (ATP) via the phosphorylation of adenosine diphosphate (ADP) follows a circadian rhythm that is entrained, or synchronized, to the 24-h light–dark cycle (22). When cyanobacteria are subjected to an abnormal light–dark cycle with an 8-h dark period during the biological day, thereby causing a misalignment between the internal circadian physiology and environment, the ATP/ADP ratio is reduced, and in turn, enzymatic reactions important both for ATP metabolism and circadian regulation are also reduced (22). In red blood cells that are dependent solely on glycolysis for ATP production, assayed NADH/NADPH ratios, and therefore ATP production, also exhibit a clear ~24-h circadian rhythm (23). In more complex systems, the circadian system and metabolism are linked at a genetic level. Gene expression in muscle, both white and brown adipose tissues and the liver have all been identified to follow a circadian rhythm in clock gene and metabolic-related gene expression (e.g. rhythmic genes include those playing key roles in oxidative phosphorylation and fat metabolism) (24,25). Additionally, the rate-limiting enzymes in many metabolic processes follow 24-h circadian rhythms (26). At a transcriptional level, many core circadian clock genes are tightly coupled with metabolism. Mice with a mutation of the CLOCK (circadian locomotor output cycles kaput) gene not only show an altered and blunted circadian rhythm in wheel running in comparison with heterozygous mice but also exhibit a decrease in energy expenditure (Fig. 1) with an increase in caloric intake and body mass (27). Orphan nuclear hormone receptors reverse erythroblastosis virus α (Rev-erbα) and receptor-related orphan receptor α, which are critical for regulation of the BMAL-1 (brain muscle Arnt-like factor) gene expression in the circadian transcriptional–translational feedback loop, are also important for regulation of lipid metabolism (28). Dys-regulation of BMAL-1 in mice leads to changes in energy and lipid homeostasis similar to that of metabolic syndrome (29). Furthermore, Rev-erbα and Rev-erbβ double knockout mice show both circadian arrythmicity in wheel running activity and a deregulation of lipid metabolism (30). Similarly, peroxisome proliferator-activated receptor gamma coactivator 1-α, the transcriptional co-activator that regulates nuclear hormone receptors, has been suggested to be the key integrator of clock and metabolic pathways because it has a robust circadian rhythm and is susceptible to both environmental and nutritional signals (31).

Less is known about the integration between energy metabolism and the circadian system in humans. Using a 41-h protocol in which wakefulness was maintained and all environmental and behavioural factors remained constant to avoid masking of a natural rhythm of energy expenditure (i.e. participant remained seated, meals were equally distributed and lights dimmed (32)), Spengler and colleagues observed a near 24-h endogenous rhythm in carbon dioxide production with a peak in the evening around 18:00 h (33). In men under a 34-h constant routine protocol, Kräuchi and colleagues observed a 24-h rhythm to energy expenditure (kcal d^{-1}) with a peak during the middle of the day and a trough between midnight and 06:00 h (34). Additional research is needed to examine the circadian timing and amplitude of energy metabolism as it relates to other factors (e.g. sleep–wake) and how chronic disruption of the circadian system may alter energy metabolism.

Insufficient and disrupted sleep and energy metabolism

Because of time demands of modern society with work, school and social schedules, insufficient sleep is common amongst both adults (35) and adolescents (36). Epidemiological evidence has produced numerous associations of short sleep and obesity (37–39). However, it would seem somewhat paradoxical that short sleep would result in weight gain, as wakefulness increases energy expenditure (10,16,18). Thus, if energy intake was stable, the increased amount of wakefulness and associated increase in energy expenditure would lead to a negative shift in energy balance and consequent weight loss over time. Indeed, findings from rodent models show sleep restriction and sleep disturbance to increase total daily energy expenditure and result in either weight loss or weight stability despite that total daily caloric intake is also increased (40,41). When sleep is restricted in humans and food is provided *ad libitum*, participants have been shown to increase their food intake far beyond the amount of calories needed to meet the energy needs of extended wakefulness and thus enter a state of positive energy balance (16,42,43). If maintained over time, this positive energy balance will result in weight gain (16,43). Furthermore, the response of increased caloric intake and subsequent weight gain during sleep restriction appears to be a consistent response within an individual under repeated exposures of sleep restriction (44). Insufficient sleep also impacts the resulting body composition during caloric

© 2017 World Obesity Federation

restriction-induced weight loss. Overweight individuals exhibit similar weight loss during caloric restriction regardless of sleep, but calorically restricted individuals under insufficient sleep conditions showed a higher loss of fat-free mass whereas calorically restricted individuals under adequate sleep conditions showed higher loss of fat mass (45). These data indicate that healthy sleep results in a healthier body composition when dieting.

Mechanistically, the causes of increased caloric intake during sleep restriction may be varied. Participants in laboratory studies have shown changes in appetitive hormones and reported increased hunger and appetite ratings on visual analogue scales (46,47), and functional magnetic resonance imaging scans have revealed increased activation of brain regions that may be sensitive to food stimulation (48,49). During extended wakefulness, there is more opportunity to eat, and changes in the timing of food intake have been observed (described next). Acute sleep restriction and controlled caloric intake lower the maximal and rhythmic 24-h levels of the satiety hormone leptin (46,50–52), insulin (53) and signalling to adipose tissue that communicates to the brain to maintain energy balance (54). Further promoting caloric intake, sleep restriction also increases circulating levels of the appetitive hormone ghrelin (46,47,51), which can predict the amount of calories an individual will consume during sleep restriction (55). However, these changes have been observed when food is maintained at levels needed for energy balance at baseline. Over-eating during sleep loss more often results in reduced ghrelin and increased leptin and peptide YY levels, which should reduce food intake (16,56,57). That excess food intake is observed despite such changes in appetitive hormones and indicates that other physiological drives, such as higher energy expenditure (16), contribute to high food intake during sleep loss.

In addition to total sleep deprivation or insufficient nightly sleep, the metabolic costs of disturbed or fragmented sleep should also be considered as they commonly occur in modern society as a result of untreated sleep problems or environmental factors such as noise, ambient temperature and drugs (e.g. recreational, caffeine, alcohol and nicotine). Fragmented sleep induced by acoustic stimulation during the biological night has been shown to increase energy expenditure concurrent with decreases in rapid eye movement (REM) and deep slow wave sleep (SWS) and increases in the lighter stage 1 sleep (15). Further, the amount of SWS has been inversely correlated with energy balance whereas REM sleep has been positively correlated with energy balance (58), although findings comparing energy expenditure levels during different sleep stages have shown small differences between stages (17,19). In a single-blind crossover design experiment in which subjects were forced to turn off an alarm intermittently while sleeping in a respiratory chamber, fragmented sleep not only led to decreases in total sleep time, SWS and REM sleep but also was associated with increases in carbohydrate oxidation, energy expenditure, exhaustion and decreases in fat oxidation (59). Taken together, fragmented sleep may lead to an increase in energy expenditure from increased wakefulness, but have a deleterious impact on metabolism afterwards.

As noted previously, sleep restriction (yoked control rats) and sleep disturbance in rodents appear to induce a higher metabolic cost (e.g. ~20% increase; Fig. 1) than that observed in sleep-restricted humans (e.g. ~5% increase) and rodents lose weight or are weight stable whereas humans gain weight under *ad libitum* feeding conditions. Thus, although rodent models recapitulate some of the energetic costs of sleep loss, the changes in energy balance are different, thus limiting direct translation to humans. Most sleep loss studies that have examined energy expenditure and energy balance in humans, however, are of short duration, and additional research is needed to examine whether there are any physiological adaptations to more chronic insufficient sleep or sleep disturbance.

Insufficient and disrupted sleep and glucose metabolism

Most of the research on sleep and glucose metabolism has focused on sleep duration as a possible mechanism leading to disrupted glucose regulation and insulin sensitivity (60–63). In a seminal study by Spiegel and colleagues, restricting sleep to a 4-h opportunity per night for six nights was found to decreased glucose clearance by 40% and the acute insulin response to glucose by 30% during a frequently sampled intravenous glucose tolerance test (IVGTT), similar to levels observed in diabetic populations (60). Similar findings of disturbed glucose regulation have been found when sleep duration was reduced to 5.5-h sleep opportunity for 14 d (64). Furthermore, reduced insulin sensitivity was observed following a 5-h sleep opportunity per night for 1 week – ~20% reduced during an IVGTT and ~11% reduced during a euglycaemic hyperinsulinemic clamp – (63) and following a 4-h sleep opportunity for one night (62). Disrupting SWS for three nights using acoustic stimuli has also been found to decrease insulin sensitivity in adults (65), although not after one night of disruption in pubertal children (66). Mechanisms for decreased glucose metabolism in response to sleep disturbance are still being uncovered. In a recent study by Broussard and colleagues, participants were studied in a within subject design of either four consecutive nights of a 4.5-h sleep opportunity or four nights of an 8.5-h sleep opportunity (67). They found that during the sleep restriction protocol, participants had elevated non-esterified free fatty acids levels during the insulin assessment, which have been documented to decrease insulin sensitivity (68). Furthermore, a higher level of non-esterified free fatty acids after sleep restriction was associated with a lower level of insulin sensitivity (67). Other proposed mechanisms such as

 © 2017 World Obesity Federation

elevated cortisol (63,64,67,69) or changes in sympathetic nervous system activity (60,67) are not as conclusive. Further work is needed to understand potential underlying mechanisms by which sleep disturbance leads to dysregulated metabolism and how such mechanisms may differ with sex, age, lifestyle and in comorbid disease populations.

Extending sleep by 1 h per night for 6 weeks in individuals who maintain habitual ~6-h short sleep schedules improved glucose metabolism (70). On a shorter timescale, 3 d of 'catch-up' sleep of 10-h sleep opportunity (71) or 3 d of 9-h sleep opportunity (72) has also been shown to improve oral glucose tolerance measures of insulin sensitivity. Further, providing 9 d of extended recovery sleep opportunities of 10-h time-in-bed to participants in the laboratory after weeks of restricted and misaligned sleep restores glucose metabolism to baseline levels in healthy individuals (73). Taken together, these results suggest that short sleep duration might be a modifiable behaviour to improve glucose metabolism and should be explored as a potential countermeasure for disease as it does not require pharmacological intervention.

Circadian misalignment and energy metabolism

As reviewed previously, the circadian system and energy metabolism are tightly integrated to promote energy intake during the day. In the modern 24-h society, however, people often choose to partake in these behaviours at inappropriate biological times. This misalignment is often accompanied with poor health. Night shift workers have a higher risk of gastrointestinal disorders, (74–76) obesity (77,78), cardiovascular disease (79) and diabetes (80,81). In rodent models, allowing mice to eat only during their habitual sleep time – thereby inducing a circadian misalignment as the mice must be awake to eat – increases body mass as compared with mice eating during habitual wake time even though caloric intake or locomotor activity was similar (82). Furthermore, Salgado-Delgado and colleagues found that simulating a shift work schedule in mice shifted feeding patterns to the 'working' hours (typical sleep phase) and increased abdominal obesity even though they did not consume more calories compared with controls (83). Restricting food intake to the typical wake phase while continuing to maintain a night shift schedule abolished these differences (83). Exposing mice to constant bright light throughout the 24 h d^{-1} resulted in an increased body mass with simultaneous reductions in glucose tolerance, again occurring in the absence of increased caloric intake or decreased activity (84). Moreover, subsequent restriction of the feeding opportunity to the typical wake phase prevented the weight gain (84). Coomans and colleagues reported that keeping mice in either a constant bright light and feeding them either a normal or high-fat chow, resulted in faster weight gain compared with mice eating high-fat or normal chow diets in a cycled light–dark environment (85). Although the mice in the aligned environment eating a high-fat chow gained weight compared with normal chow, the group eating the high-fat chow in the misaligned environment displayed additive effects and the highest weight gain (85).

In humans, sleeping during the biological day and wakefulness and eating during the biological night in a shift work model decreased 24-h energy expenditure (17). The latter may help to explain a potential contributing mechanism as to why the rodent models and shift workers eating during the biological night gain weight in the absence of higher caloric intake or energy expenditure differences. In a 6-d protocol to simulate the transition to and subsequent nights of shift work, we studied participants in a whole room indirect calorimeter for minute-by-minute energy expenditure while controlling for activity and caloric intake (17). Similar to studies of sleep restriction, we found that the transition to shift work with extended wakefulness resulted in an increased energy expenditure, but that the sleeping energy expenditure prior to the night shifts and 24-h energy expenditure of the night shifts were decreased as compared with the day shift (Fig. 1) (17). If maintained for extended durations and if caloric intake was not restricted, this may place individuals into a positive energy balance and weight gain over time. Further contributing to the decreased energy expenditure, we also found a decreased thermic effect of food in response to dinner on the first night shift when the meal was consumed at ~22:30 h, consistent with other findings of a decreased thermic effect of food when a meal is consumed at night as compared with the day (86,87). As shift workers self-report eating a similar amount of calories as their daytime counterparts (2), potential countermeasures may need to focus on combating the decreases in energy expenditure to improve health and decrease the weight gains observed in shift work. For example, shift workers may require more physical activity associated energy expenditure to maintain energy balance and metabolic health as compared with day workers. Also, it is unknown if there are more effective circadian times for exercise in shift workers with respect to metabolic health benefits.

Circadian misalignment and timing of food intake

In both shift working and non-shift working populations, the decreased energy expenditure in response to a meal at night may play a very important role in weight gain and adverse metabolic health. Recent evidence has emerged indicating that circadian-driven hunger ratings on a visual analogue scale are highest at ~20:00 h and lowest at ~07:50 h (88), potentially a physiological drive to prepare for the nightly fast. However, as food is easily accessible, eating at night is associated with unwanted weight gain. Related, in

© 2017 World Obesity Federation

a study of early and late timing of caloric intake, researchers found that eating an 'early' lunch (before 15:00 h) rather than a 'late' lunch (after 15:00 h) was associated with greater weight loss during active dieting with no differences between the two groups estimated energy expenditure, caloric intake or sleep duration (89). Participants in the 'late' lunch group were also more likely to skip breakfast and be considered evening types, potentially leading to post lunch meals occurring later in the evening than 'early' lunch eaters (89). The timing of a 'late' or 'early' main meal has also been found to be associated with weight loss effectiveness after bariatric surgery, with early eaters showing more weight loss success than later eaters (90). Baron and colleagues have also found that high caloric intake past 20:00 h predicts an individual's body mass index, even when controlling for sleep timing and duration (91). Night eating syndrome, a condition in which individuals consume a significant portion of their daily food intake late into the evening and at night, is associated with a higher body mass index and binge eating tendencies (92).

In contrast, restricting the timing of food intake in humans to only the daytime hours results in lower food intake and weight loss. In a recent study by Gill and colleagues, participants were asked to restrict the timing of their caloric consumption between the hours of 06:00 and 18:00 h for up to 1 year and document the timing of meals on a mobile phone application (93). Similar to the rodent models of time-restricted caloric intake to the biological day, they found a significant decrease in weight after 16 weeks of intervention. Furthermore, participants reported concurrent increases in overall energy levels and morning energy levels and decreases in hunger at bedtime (93).

Sleep restriction may also play a role in the promotion of eating at an adverse circadian time. In a 14- to 15-d randomized cross-over design study in which subjects were either given a 5- or 9-h sleep opportunity for 5 d with *ad libitum* food access, our group found that subjects in the sleep-restricted 5-h group not only gained 0.82 kg on average despite expending ~5% more energy than the 9-h condition but also consumed more calories in post dinner snacks than any other individual meal (16). Furthermore, Nedeltcheva and colleagues found that restricting subjects to 5.5 h of sleep opportunity as opposed to 8 h of sleep opportunity resulted in increased snack consumption between the hours of 19:00 and 07:00 h (56). Restricting sleep to only 4-h time-in bed also increases the consumption of night-time snacking (43). Whether restricting food intake to the biological daytime during sleep loss prevents weight gain is unknown. Future research is needed to disentangle the timing of food consumption in relation to circadian timing in free-living conditions in order to identify mechanisms that promote metabolic dys-regulations and adverse health.

Circadian misalignment and glucose metabolism

The relationship between circadian misalignment and impaired glucose metabolism has also been of high interest for metabolic health research. In a protocol designed to distribute all activities and behaviours evenly across all phases of the internal circadian cycle (forced desynchrony protocol (94)), Scheer and colleagues found that when participants were awake and ate 12 h out of phase from their habitual times, there was a 6% increase in glucose concentrations despite a 22% increase in insulin levels across the behavioural cycle (95). Using a similar protocol but also including sleep restriction equivalent to 5.6 h per 24 h d^{-1}, Buxton and colleagues found that glucose levels increased with a concomitant decrease in insulin levels (73). We have also found that combining sleep restriction with circadian misalignment decreases early morning insulin sensitivity in healthy lean individuals by ~20% as measured by both IVGTT and oral glucose tolerance tests (72). These findings have importance as disturbed sleep and circadian misalignment are difficult to dissociate in real-world settings as the systems are highly integrated (96). In a study of healthy lean adults, Leproult and colleagues imposed short sleep duration under circadian aligned and misaligned conditions and found that circadian misalignment combined with short sleep decreased IVGTT insulin sensitivity to a greater extent than short sleep duration alone (97). Decreased glucose metabolism in response to mixed meal testing was also observed in the evening hours when controlling for sleep duration and behavioural factors such as activity and feeding/fasting (98), and does not appear to adapt to repeated exposures to circadian misalignment, as common in chronic overnight shift workers (99).

Physiological mechanisms of decreased glucose metabolism during circadian misalignment are being explored. A key difference in circadian physiology between diurnal species such as humans, and nocturnal species such as many rodents, is the difference in timing of food intake with respect to the circadian-driven hormone melatonin, which communicates the time of solar darkness. Human metabolic physiology is prepared for food intake when melatonin levels are low, whereas nocturnal rodent metabolic physiology is prepared for food intake when endogenous melatonin levels are high. This difference in melatonin and wake–sleep/feeding–fasting timing between diurnal and nocturnal species has implications for understanding the role of melatonin (100,101) and of melatonin receptor polymorphisms (102–105) in metabolic disease risk. In humans obtaining sufficient sleep, melatonin levels typically increase ~2 h prior to habitual sleep time and stay elevated throughout the night. The timing of melatonin can shift depending on light exposure (106,107); however, during misalignment such as night shift work or rapid transmeridian jet travel, the melatonin rhythm is unable to fully or quickly adjust

 © 2017 World Obesity Federation

(108). When melatonin is administered to young (101) or postmenopausal (100) women in the morning or evening hours when endogenous levels are not high, glucose tolerance and insulin sensitivity become decreased, potentially providing evidence that the presence of melatonin may be a mechanisms for impaired glucose metabolism. Furthermore, recent work has identified a specific genetic variant in the melatonin receptor, common in up to 51% of individuals with European ancestry and a known risk factor for diabetes (104), as further impairing glucose metabolism when melatonin is administered (103). This genetic variant may also predispose individuals for a longer circadian night (109), increasing the potential time of circadian misalignment and decreased glucose metabolism.

Conclusions

There is strong evidence from both animal and human models to suggest that disrupted sleep and circadian misalignment contribute to weight gain, obesity and adverse metabolic health. Although diet and activity are recognized as primary factors contributing to poor metabolic health, sleep duration, quality and the circadian timing of food intake need to be considered as contributing factors underlying metabolic disease. On the basis of the findings reviewed previously, a calorie is no longer a calorie because metabolic outcomes are dependent on the circadian timing of food intake (e.g. greater weight gain in nocturnal rodent models when food intake is restricted to the typical sleep phase and altered glucose and energy metabolism when food intake occurs during the biological night in humans). Additional research is needed to understand mechanisms by which sleep and circadian disruption contribute to metabolic dys-regulation/alterations and to determine the potential of improving sleep and circadian rhythms and the scheduling of daily activities (e.g. food intake) to promote immediate and long-term metabolic health.

Acknowledgements

This work was supported by the National Institutes of Health grants R21 DK092624 (K.P.W.), R01 HL109706 (K.P.W.), F32DK107146 (A.W.M.) and T32HL007901 (A.W.M.).

Conflict of interest statement

A.W.M. reports no conflicts; K.P.W.: funding: NIH, Office of Naval Research, Philips Inc., Torvec Inc.; consulting fees from or served as a paid member of scientific advisory boards for NIH, Torvec Inc.; speaker honorarium fees from American College of Chest Physicians, The Obesity Society, and Obesity Medicine Association; travel reimbursement for invited talk by the Swiss Obesity Society – Fribourg Obesity Research Conference.

References

1. Stenvinkel P. Obesity – a disease with many aetiologies disguised in the same oversized phenotype: has the overeating theory failed? *Nephrol Dial Transplant* 2015; **10**: 1656–1664.
2. Markwald RR, Wright KP Jr. Circadian misalignment and sleep disruption in shift work: implications for fatigue and risk of weight gain and obesity. *Sleep Loss and Obesity* Springer 2012: 101–118.
3. Depner CM, Stothard ER, Wright KP Jr. Metabolic consequences of sleep and circadian disorders. *Curr Diab Rep* 2014; **14**: 507.
4. Arble DM, Bass J, Behn CD *et al.* Impact of sleep and circadian disruption on energy balance and diabetes: a summary of workshop discussions. *Sleep* 2015; **38**: 1849–1860.
5. Schoenborn CA, Adams PE. Health behaviors of adults: United States, 2005–2007. *Vital and health statistics Series 10, Data from the National Health Survey*. 2010: 1-132.
6. Hirshkowitz M, Whiton K, Albert SM *et al.* National Sleep Foundation's sleep time duration recommendations: methodology and results summary. *Sleep Health* 2015; **1**: 40–43.
7. Wright KP Jr, McHill AW, Birks BR, Griffen BR, Rusterholz T, Chinoy ED. Entrainment of the human circadian clock to the natural light–dark cycle. *Curr Biol* 2013; **23**: 1554–1558.
8. McMenamin TM. A time to work: recent trends in shift work and flexible schedules. *Monthly LabRev* 2007; **130**: 3.
9. Ogden CL, Carroll MD, Kit BK, Flegal KM. Prevalence of obesity among adults: United States, 2011–2012. *NCHS Data Brief* 2013; **131**: 1–8.
10. Rechtschaffen A, Gilliland MA, Bergmann BM, Winter JB. Physiological correlates of prolonged sleep deprivation in rats. *Science* 1983; **221**: 182–184.
11. Everson CA, Bergmann BM, Rechtschaffen A. Sleep deprivation in the rat: III. Total sleep deprivation. *Sleep* 1989; **12**: 13–21.
12. Bergmann BM, Everson CA, Kushida CA *et al.* Sleep deprivation in the rat: V. Energy use and mediation. *Sleep* 1989; **12**: 31–41.
13. Kreider MB, Buskirk ER, Bass DE. Oxygen consumption and body temperatures during the night. *J Appl Physiol* 1958; **12**: 361–366.
14. Fraser G, Trinder J, Colrain IM, Montgomery I. Effect of sleep and circadian cycle on sleep period energy expenditure. *J Appl Physiol* 1989; **66**: 830–836.
15. Bonnet MH, Berry RB, Arand DL. Metabolism during normal, fragmented, and recovery sleep. *J Appl Physiol* 1991; **71**: 1112–1118.
16. Markwald RR, Melanson EL, Smith MR *et al.* Impact of insufficient sleep on total daily energy expenditure, food intake, and weight gain. *Proc Natl Acad Sci U S A* 2013; **110**: 5695–7500.
17. McHill A, Melanson E, Higgins J *et al.* Impact of circadian misalignment on energy metabolism during simulated shiftwork. *Proc Natl Acad Sci U S A* 2014; **111**: 17302–17307.
18. Shechter A, Rising R, Albu JB, St-Onge MP. Experimental sleep curtailment causes wake-dependent increases in 24-h energy expenditure as measured by whole-room indirect calorimetry. *Am J Clin Nutr* 2013; **98**: 1433–1439.
19. Jung CM, Melanson EL, Frydendall EJ, Perreault L, Eckel RH, Wright KP. Energy expenditure during sleep, sleep deprivation and sleep following sleep deprivation in adult humans. *J Physiol* 2011; **589**: 235–244.
20. Benedict C, Hallschmid M, Lassen A *et al.* Acute sleep deprivation reduces energy expenditure in healthy men. *Am J Clin Nutr* 2011; **93**: 1229–1236.
21. Schmidt MH. The energy allocation function of sleep: a unifying theory of sleep, torpor, and continuous wakefulness. *Neurosci Biobehav Rev* 2014; **47**: 122–153.

© 2017 World Obesity Federation

22. Rust MJ, Golden SS, O'Shea EK. Light-driven changes in energy metabolism directly entrain the cyanobacterial circadian oscillator. *Science* 2011; **331**: 220–223.
23. O'Neill JS, Reddy AB. Circadian clocks in human red blood cells. *Nature* 2011; **469**: 498–503.
24. Yang X, Downes M, Yu RT *et al.* Nuclear receptor expression links the circadian clock to metabolism. *Cell* 2006; **126**: 801–810.
25. Green CB, Takahashi JS, Bass J. The meter of metabolism. *Cell* 2008; **134**: 728–742.
26. Panda S, Antoch MP, Miller BH *et al.* Coordinated transcription of key pathways in the mouse by the circadian clock. *Cell* 2002; **109**: 307–320.
27. Turek FW, Joshu C, Kohsaka A *et al.* Obesity and metabolic syndrome in circadian clock mutant mice. *Science* 2005; **308**: 1043–1045.
28. Fontaine C, Staels B. The orphan nuclear receptor Rev-erb [alpha]: a transcriptional link between circadian rhythmicity and cardiometabolic disease. *Curr Opin Lipidol* 2007; **18**: 141–146.
29. Shimba S, Ogawa T, Hitosugi S *et al.* Deficient of a clock gene, brain and muscle Arnt-like protein-1 (BMAL1), induces dyslipidemia and ectopic fat formation. *PLoS One* 2011; **6**: e25231.
30. Cho H, Zhao X, Hatori M *et al.* Regulation of circadian behaviour and metabolism by REV-erb-alpha and REV-erb-beta. *Nature* 2012; **485**: 123–127.
31. Lin JD, Liu C, Li S. Integration of energy metabolism and the mammalian clock. *Cell Cycle* 2008; **7**: 453–457.
32. Mills JN, Minors DS, Waterhouse JM. Adaptation to abrupt time shifts of the oscillator(s) controlling human circadian rhythms. *J Physiol* 1978; **285**: 455–470.
33. Spengler CM, Czeisler CA, Shea SA. An endogenous circadian rhythm of respiratory control in humans. *J Physiol* 2000; **526**: 683–694.
34. Kräuchi K, Wirz-Justice A. Circadian rhythm of heat production, heart rate, and skin and core temperature under unmasking conditions in men. *Am J Phys* 1994; **267**: R819–R829.
35. Krueger PM, Friedman EM. Sleep duration in the United States: a cross-sectional population-based study. *Am J Epidemiol* 2009; **169**: 1052–1063.
36. Carskadon MA. Sleep in adolescents: the perfect storm. *Pediatr Clin N Am* 2011; **58**: 637–647.
37. Patel SR, Hu FB. Short sleep duration and weight gain: a systematic review. *Obesity* 2008; **16**: 643–653.
38. Bo S, Ciccone G, Durazzo M *et al.* Contributors to the obesity and hyperglycemia epidemics. A prospective study in a population-based cohort. *Int J Obes* 2011; **35**: 1442–1449.
39. Mozaffarian D, Hao T, Rimm EB, Willett WC, HFB. Changes in diet and lifestyle and long-term weight gain in women and men. *New Engl J Med* 2011; **364**: 2392–2404.
40. Everson CA, Szabo A. Repeated exposure to severely limited sleep results in distinctive and persistent physiological imbalances in rats. *PLoS One* 2011; **6**: e22987.
41. Vetrivelan R, Fuller PM, Yokota S, Lu J, Saper CB. Metabolic effects of chronic sleep restriction in rats. *Sleep* 2012; **35**: 1511–1520.
42. Brondel L, Romer MA, Nougues PM, Touyarou P, Davenne D. Acute partial sleep deprivation increases food intake in healthy men. *Am J Clin Nutr* 2010; **91**: 1550–1559.
43. Spaeth AM, Dinges DF, Goel N. Effects of experimental sleep restriction on weight gain, caloric intake, and meal timing in healthy adults. *Sleep* 2013; **36**: 981–990.
44. Spaeth AM, Dinges DF, Goel N. Phenotypic vulnerability of energy balance responses to sleep loss in healthy adults. *Sci Rep* 2015; **5**: 14920.
45. Nedeltcheva AV, Kilkus JM, Imperial J, Schoeller DA, Penev PD. Insufficient sleep undermines dietary efforts to reduce adiposity. *Ann Intern Med* 2010; **153**: 435–441.
46. Spiegel K, Tasali E, Penev P, Van Cauter E. Brief communication: sleep curtailment in healthy young men is associated with decreased leptin levels, elevated ghrelin levels, and increased hunger and appetite. *Ann Intern Med* 2004; **141**: 846–850.
47. Schmid SM, Hallschmid M, Jauch-Chara K, Born J, Schultes B. A single night of sleep deprivation increases ghrelin levels and feelings of hunger in normal-weight healthy men. *J Sleep Res* 2008; **17**: 331–334.
48. St-Onge MP, McReynolds A, Trivedi ZB, Roberts AL, Sy M, Hirsch J. Sleep restriction leads to increased activation of brain regions sensitive to food stimuli. *Am J Clin Nutr* 2012; **95**: 818–824.
49. Greer SM, Goldstein AN, Walker MP. The impact of sleep deprivation on food desire in the human brain. *Nat Commun* 2013; **4**: 2259.
50. Spiegel K, Leproult R, L'Hermite-Baleriaux M, Copinschi G, Penev PD, Van Cauter E. Leptin levels are dependent on sleep duration: relationships with sympathovagal balance, carbohydrate regulation, cortisol, and thyrotropin. *J Clin Endocrinol Metab* 2004; **89**: 5762–5771.
51. Taheri S, Lin L, Austin D, Young T, Mignot E. Short sleep duration is associated with reduced leptin, elevated ghrelin, and increased body mass index. *PLoS Med* 2004; **1**: e62.
52. Chaput JP, Després JP, Bouchard C, Tremblay A. Short sleep duration is associated with reduced leptin levels and increased adiposity: results from the Quebec family study. *Obesity* 2007; **15**: 253–261.
53. Nedeltcheva AV, Imperial JG, Penev PD. Effects of sleep restriction on glucose control and insulin secretion during diet-induced weight loss. *Obesity* 2012; **20**: 1379–1386.
54. Broussard JL, Ehrmann DA, Van Cauter E, Tasali E, Brady MJ. Impaired insulin signaling in human adipocytes after experimental sleep restriction: a randomized, crossover study. *Ann Intern Med* 2012; **157**: 549–557.
55. Broussard JL, Kilkus JM, Delebecque F *et al.* Elevated ghrelin predicts food intake during experimental sleep restriction. *Obesity* 2016; **24**: 132–138.
56. Nedeltcheva AV, Kilkus JM, Imperial J, Kasza K, Schoeller DA, Penev PD. Sleep curtailment is accompanied by increased intake of calories from snacks. *Am J Clin Nutr* 2009; **89**: 126–133.
57. St-Onge MP, O'Keeffe M, Roberts AL, RoyChoudhury A, Laferrere B. Short sleep duration, glucose dysregulation and hormonal regulation of appetite in men and women. *Sleep* 2012; **35**: 1503–1510.
58. Rutters F, Gonnissen HK, Hursel R, Lemmens SG, Martens EA, Westerterp-Plantenga MS. Distinct associations between energy balance and the sleep characteristics slow wave sleep and rapid eye movement sleep. *Int J Obes* 2012; **36**: 1346–1352.
59. Hursel R, Rutters F, Gonnissen HK, Martens EA, Westerterp-Plantenga MS. Effects of sleep fragmentation in healthy men on energy expenditure, substrate oxidation, physical activity, and exhaustion measured over 48 h in a respiratory chamber. *Am J Clin Nutr* 2011; **94**: 804–808.
60. Spiegel K, Leproult R, Van Cauter E. Impact of sleep debt on metabolic and endocrine function. *Lancet* 1999; **354**: 1435–1439.
61. Spiegel K, Knutson K, Leproult R, Tasali E, Van Cauter E. Sleep loss: a novel risk factor for insulin resistance and type 2 diabetes. *J Appl Physiol* 2005; **99**: 2008–2019.
62. Donga E, van Dijk M, van Dijk JG *et al.* A single night of partial sleep deprivation induces insulin resistance in multiple metabolic pathways in healthy subjects. *J Clin Endocrinol Metab* 2010; **95**: 2963–2968.

 © 2017 World Obesity Federation

63. Buxton OM, Pavlova M, Reid EW, Wang W, Simonson DC, Adler GK. Sleep restriction for 1 week reduces insulin sensitivity in healthy men. *Diabetes* 2010; **59**: 2126–2133.
64. Nedeltcheva AV, Kessler L, Imperial J, Penev PD. Exposure to recurrent sleep restriction in the setting of high caloric intake and physical inactivity results in increased insulin resistance and reduced glucose tolerance. *J Clin Endocrinol Metab* 2009; **94**: 3242–3250.
65. Tasali E, Leproult R, Ehrmann DA, Van Cauter E. Slow-wave sleep and the risk of type 2 diabetes in humans. *Proc Natl Acad Sci U S A* 2008; **105**: 1044–1049.
66. Shaw ND, McHill AW, Schiavon M *et al.* Effect of slow-wave sleep disruption on metabolic parameters in adolescents. *Sleep* 2016; **39**: 1591–1599.
67. Broussard JL, Chapotot F, Abraham V *et al.* Sleep restriction increases free fatty acids in healthy men. *Diabetologia* 2015; **58**: 791–798.
68. Boden G, Lebed B, Schatz M, Homko C, Lemieux S. Effects of acute changes of plasma free fatty acids on intramyocellular fat content and insulin resistance in healthy subjects. *Diabetes* 2001; **50**: 1612–1617.
69. Gonzalez-Ortiz M, Martinez-Abundis E, Balcazar-Munoz BR, Pascoe-Gonzalez S. Effect of sleep deprivation on insulin sensitivity and cortisol concentration in healthy subjects. *Diabetes Nutr Metab* 2000; **13**: 80–83.
70. Leproult R, Deliens G, Gilson M, Peigneux P. Beneficial impact of sleep extension on fasting insulin sensitivity in adults with habitual sleep restriction. *Sleep* 2015; **38**: 707–715.
71. Killick R, Hoyos CM, Melehan KL, Dungan GC 2nd, Poh J, Liu PY. Metabolic and hormonal effects of 'catch-up' sleep in men with chronic, repetitive, lifestyle-driven sleep restriction. *Clin Endocrinol* 2015; **83**: 498–507.
72. Eckel RH, Depner CM, Perreault L *et al.* Morning circadian misalignment during short sleep duration impacts insulin sensitivity. *Curr Biol* 2015; **25**: 3004–3010.
73. Buxton OM, Cain SW, O'Connor SP *et al.* Adverse metabolic consequences in humans of prolonged sleep restriction combined with circadian disruption. *Sci Transl Med* 2012; **4** .129ra43
74. Segawa K, Nakazawa S, Tsukamoto Y *et al.* Peptic ulcer is prevalent among shift workers. *Dig Dis Sci* 1987; **32**: 449–453.
75. Knutsson A. Health disorders of shift workers. *Occup Med* 2003; **53**: 103–108.
76. Drake CL, Roehrs T, Richardson G, Walsh JK, Roth T. Shift work sleep disorder: prevalence and consequences beyond that of symptomatic day workers. *Sleep* 2004; **27**: 1453–1462.
77. Di Lorenzo L, De Pergola G, Zocchetti C *et al.* Effect of shift work on body mass index: results of a study performed in 319 glucose-tolerant men working in a Southern Italian industry. *Int J Obes* 2003; **27**: 1353–1358.
78. Antunes LC, Levandovski R, Dantas G, Caumo W, Hidalgo MP. Obesity and shift work: chronobiological aspects. *Nutr Res Rev* 2010; **23**: 155–168.
79. Vetter C, Devore EE, Wegrzyn LR *et al.* Association between rotating night shift work and risk of coronary heart disease among women. *JAMA* 2016; **315**: 1726–1734.
80. Pan A, Schernhammer ES, Sun Q, HFB. Rotating night shift work and risk of type 2 diabetes: two prospective cohort studies in women. *PLoS Med* 2011; **8**: e1001141.
81. Guo Y, Liu Y, Huang X *et al.* The effects of shift work on sleeping quality, hypertension and diabetes in retired workers. *PLoS One* 2013; **8**: e71107.
82. Arble DM, Bass J, Laposky AD, Vitaterna MH, Turek FW. Circadian timing of food intake contributes to weight gain. *Obesity* 2009; **17**: 2100–2102.
83. Salgado-Delgado R, Angeles-Castellanos M, Saderi N, Buijs RM, Escobar C. Food intake during the normal activity phase prevents obesity and circadian desynchrony in a rat model of night work. *Endocrinology* 2010; **151**: 1019–1029.
84. Fonken LK, Workman JL, Walton JC *et al.* Light at night increases body mass by shifting the time of food intake. *Proc Natl Acad Sci U S A* 2010; **107**: 18664–18669.
85. Coomans CP, van den Berg SA, Houben T *et al.* Detrimental effects of constant light exposure and high-fat diet on circadian energy metabolism and insulin sensitivity. *FASEB J* 2013; **27**: 1721–1732.
86. Romon M, Edme J-L, Boulenguez C, Lescroart J-L, Frimat P. Circadian variation of diet-induced thermogenesis. *Am J Clin Nutr* 1993; **57**: 476–480.
87. Morris CJ, Garcia JI, Myers S, Yang JN, Trienekens N, Scheer FA. The human circadian system has a dominating role in causing the morning/evening difference in diet-induced thermogenesis. *Obesity* 2015; **23**: 2053–2058.
88. Scheer FA, Morris CJ, Shea SA. The internal circadian clock increases hunger and appetite in the evening independent of food intake and other behaviors. *Obesity* 2013; **21**: 421–423.
89. Garaulet M, Gomez-Abellan P, Alburquerque-Bejar JJ, Lee YC, Ordovas JM, Scheer FA. Timing of food intake predicts weight loss effectiveness. *Int J Obes* 2013; **37**: 604–611.
90. Ruiz-Lozano T, Vidal J, de Hollanda A, Scheer FA, Garaulet M, Izquierdo-Pulido M. Timing of food intake is associated with weight loss evolution in severe obese patients after bariatric surgery. *Clin Nutr* 2016. In Press
91. Baron KG, Reid KJ, Kern AS, Zee PC. Role of sleep timing in caloric intake and BMI. *Obesity* 2011; **19**: 1374–1381.
92. Colles SL, Dixon JB, O'Brien PE. Night eating syndrome and nocturnal snacking: association with obesity, binge eating and psychological distress. *Int J Obes* 2007; **31**: 1722–1730.
93. Gill S, Panda SA. Smartphone app reveals erratic diurnal eating patterns in humans that can be modulated for health benefits. *Cell Metab* 2015; **22**: 789–798.
94. Kleitman N. Sleep and Wakefulness as Alternating Phases in the Cycle of Existence. University of Chicago Press: Chicago Illinois, 1939.
95. Scheer FA, Hilton MF, Mantzoros CS, Shea SA. Adverse metabolic and cardiovascular consequences of circadian misalignment. *Proc Natl Acad Sci U S A* 2009; **106**: 4453–4458.
96. Czeisler CA, Weitzman E, Moore-Ede MC, Zimmerman JC, Knauer RS. Human sleep: its duration and organization depend on its circadian phase. *Science* 1980; **210**: 1264–1267.
97. Leproult R, Holmbäck U, Van Cauter E. Circadian misalignment augments markers of insulin resistance and inflammation, independently of sleep loss. *Diabetes* 2014; **63**: 1860–1869.
98. Morris CJ, Yang JN, Garcia JI *et al.* Endogenous circadian system and circadian misalignment impact glucose tolerance via separate mechanisms in humans. *Proc Natl Acad Sci* 2015; **112**: E2225–E2234.
99. Morris CJ, Purvis TE, Mistretta J, Scheer FA. Effects of the internal circadian system and circadian misalignment on glucose tolerance in chronic shift workers. *J Clin Endocrinol Metab* 2016; **101**: 1066–1074.
100. Cagnacci A, Arangino S, Renzi A *et al.* Influence of melatonin administration on glucose tolerance and insulin sensitivity of postmenopausal women. *Clin Endocrinol* 2001; **54**: 339–346.
101. Rubio-Sastre P, Scheer FA, Gomez-Abellan P, Madrid JA, Garaulet M. Acute melatonin administration in humans impairs glucose tolerance in both the morning and evening. *Sleep* 2014; **37**: 1715–1719.

© 2017 World Obesity Federation

102. Prokopenko I, Langenberg C, Florez JC *et al.* Variants in MTNR1B influence fasting glucose levels. *Nat Genet* 2009; **41**: 77–81.
103. Garaulet M, Gomez-Abellan P, Rubio-Sastre P, Madrid JA, Saxena R, Scheer FA. Common type 2 diabetes risk variant in MTNR1B worsens the deleterious effect of melatonin on glucose tolerance in humans. *Metabolism* 2015; **64**: 1650–1657.
104. Lyssenko V, Nagorny CL, Erdos MR *et al.* Common variant in MTNR1B associated with increased risk of type 2 diabetes and impaired early insulin secretion. *Nat Genet* 2009; **41**: 82–88.
105. Bouatia-Naji N, Bonnefond A, Cavalcanti-Proenca C *et al.* A variant near MTNR1B is associated with increased fasting plasma glucose levels and type 2 diabetes risk. *Nat Genet* 2009; **41**: 89–94.
106. Czeisler CA, Allan JS, Strogatz SH *et al.* Bright light resets the human circadian pacemaker independent of the timing of the sleep–wake cycle. *Science* 1986; **233**: 667–671.
107. Khalsa SBS, Jewett ME, Cajochen C, Czeisler CA. A phase response curve to single bright light pulses in human subjects. *J Physiol* 2003; **549**: 945–952.
108. Folkard S. Do permanent night workers show circadian adjustment? A review based on the endogenous melatonin rhythm. *Chronobiol Int* 2008; **25**: 215–224.
109. Lane JM, Chang AM, Bjonnes AC *et al.* Impact of common diabetes risk variant in MTNR1B on sleep, circadian and melatonin physiology. *Diabetes* 2016; **65**: 1741–1751.
110. Everson CA, Gilliland MA, Kushida CA *et al.* Sleep deprivation in the rat: IX. Recovery. *Sleep* 1989; **12**: 60–67.

 © 2017 World Obesity Federation